JECRC University, Jaipur
Department of Physics
B. Tech Ist Year – Ist Semester July-Dec 2025-26
Applied Physics (DPH001B)
Assignment-II

A. Very Short Answer Questions **(10)**

1. Write two shortcomings of Drude Lorentz theory for free electrons in metals? (2)
2. Which distribution do free electrons follow in classical Lorentz Drude theory? (2)
3. Write the correction factor to consider the Pauli Exclusion Principle while calculating density of available states. (2)
4. Define Fermi Gas. (2)
5. While formation of bands in solids which bands are affected most. (2)

B. **Answer the following briefly** **(21)**

1. Write the expression for Fermi Dirac Distribution function. Plot the function for different cases $E< E_F$, $E= E_F$ and $E> E_F$ when $T=0^oK$ and $T>0^oK$. What is value of Fermi Dirac distribution function when $E=E_F$ and $T=0^oK$. **(7)**
2. If the Fermi energy for copper is 4eV, calculate the density of electrons in copper at 0^o K. **(7)**
3. Calculate the Fermi energy in copper assuming that each copper atom contributes one free electron to the electron gas. Given density of copper $8.94x10^3$ kg/m^3 and atomic mass is 63.5u? **(7)**

C. **Answer the followings explicitly.** **(33)**

1. Write the assumptions of classical free electron theory. What was its success and what was its failure? Write the corrections done Sommerfeld to overcome the shortcomings of classical free electron theory. **(11)**
2. Using the Sommerfeld free electrons theory derive the expression for density of available states for the free electrons in solids. **(11)**
3. Taking the example of Sodium ‘Na’ explain the formation of energy bands in solids. **(11)**